

Table 1: *Material Properties*

Material	$\alpha \left(\frac{\mu\text{m}}{\text{m} \cdot ^\circ\text{C}} \right)$	$k \left(\frac{\text{W}}{\text{m} \cdot ^\circ\text{C}} \right)$	$E \text{ (GPa)}$	ν	$c_p \left(\frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}} \right)$
Aluminum 6061	22.85	167	68.95	—	962
ALLVAR	−30	8.88	55	—	—
Invar 36	1.3	10.5	141	—	515
Inconel 718	13.2	—	200	—	435
316 Stainless Steel	—	—	207.8	0.282	—